Project Management Plan

SQRT (Sub-zero Quantitative Research Team)

Bruce Ritter, Elyana Abdallah, Javier Ayuso, Caimin Keavney, Orlaith Ni Dhuill

Document History

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Key Reference Documents

1. SQT-PDR-MDD-0
2. SQT-PDR-SEP-0
3. SQT-PDR-DJF-0
4. UCD-DRD-2026-0
5. UCD-SCD-2026-0

1 Executive Summary

This document will provide an overview of the management plan for the SQRT mission, which encompasses the organisational structure, the work breakdown structure, the project timeline, and the budget and procurement strategy. The SQRT mission will investigate the behaviour of water exposed to a low pressure, low temperature environment using a demonstration at high altitude. This demonstration will be done by a TupperSat payload flying on a high-altitude balloon. The project management plan outlined in this document will facilitate the completion of the mission goals and requirements in a timely and cost-effective manner.

The SQRT team includes five members who each lead a section of the mission. Leadership roles in the team are divided across the science, structure, AIV (assembly, integration, and verification), software, and project management aspects of the project. Team members will work collaboratively on all aspects of the project with team leads having the final responsibility. The work required to satisfy the project requirements is organized into work packages. While work on the work packages will be similarly collaborative, one team member is assigned ownership over completion of each work package. The work breakdown structure and work packages are detailed in Sec. 4 of this document.

The timeline for this project follows six phases (Phase 0 - Phase 5). In order the phases cover: defining a mission concept, initial design, procurement, prototyping, and software development, manufacturing and testing, the mission operation, and post-flight analysis. Sec. 5 includes a more in depth timeline relating the mission schedule to work packages and key dates.

The budget for this project comes entirely from University College Dublin funding. The expected expenses for the SQRT project are outlined in Sec. 6.2 and total €82.71, which leaves a significant margin for unforeseen expenses.

The organisational structure, work breakdown structure, and budget outlined in this document will allow the SQRT team to complete all mission requirements effectively within the set timeline while remaining within the budget.

2 Project Overview

Mission Statement: The SQRT mission will produce and observe the flash evaporation of water via exposure to a low pressure and low temperature environment to answer the question of how water plumes on icy moons thermally evolve.

This investigation into the thermal mechanisms behind cryovolcanic plumes will be done using a high-altitude demonstration. The design of the project includes a mechanism to expose liquid water to another container kept close to the temperature and pressure of the outside near-space environment at a set altitude. The temperature of the water sample will be measured overtime via a thermal camera as it undergoes a predicted flash. A more detailed description of the mission design can be found in document SQT-PDR-MDD-0. The project management plan is affected by the mission objectives, mission requirements (Level 1), and technical requirements (Level 2), which are outlined in SQT-PDR-MDD-0 and SQT-PDR-SEP-0. These requirements affect the overall mission design. Additionally, the budget and procurement strategy is affected by the component choices in the design overview in SQT-PDR-DJF-0. The scheduling of work packages is influenced by the procurement strategy, the Level 1 and Level 2 requirements, and key milestones outlined in Sec. 5.

2.1 Mission Objectives

The project management plan will facilitate the timely completion of the following SQRT mission objectives:

MO1 Fly and recover a TupperSat containing an experimental scientific or engineering/technology demonstration payload with potential application to space science.

MO2 SQRT will attempt to reproduce and observe the flash evaporation process of water exposed to near-space conditions.

3 Organisational Structure

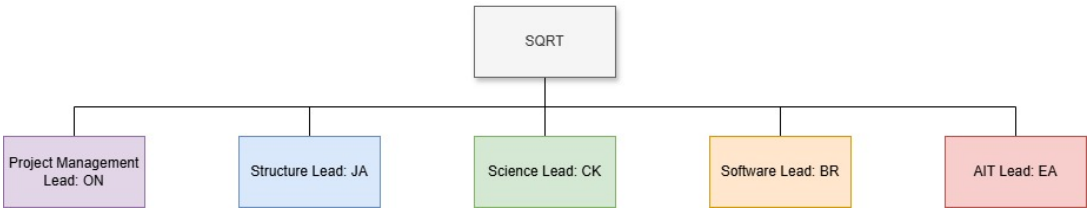


Figure 1: SQRT Organigram

The SQRT project is overseen by five lead engineers: a project manager, a science lead, a structure lead, a software lead, and an AIV lead. These lead engineers will oversee key parts of the SQRT project, based on their background. These sections correspond to five of the main work packages in the work breakdown structure of this mission. However, the SQRT team will collaborate on all aspects of the mission so team leads will not hold full responsibility over all sub work packages within the section they lead. Responsibility for individual work packages are detailed in in Sec. 4: Work Breakdown Structure.

3.1 Key Responsibilities

In addition to work packages, team members are responsible for authoring and reviewing mission documents. The key responsibilities and authored documents are listed by team member in Table 1.

Table 1: SQRT Key Responsibilities

Team Member	Key Responsibilities	Document Authorship
Elyana Abdallah	Leading AIV	SQT-PDR Final Review, other documents TBA
Javier Ayuso Vazquez	Leading Structure Development	SQRT-PDR-DJF Author, other documents TBA
Caimin Keavney	Leading Science	SQT-PDR-SEP Author, other documents TBA
Orlaith Ni Dhuill	Project Manager	SQT-PDR-MDD Author, other documents TBA
Bruce Ritter	Leading Software Development	SQt-PDR-PMP Author, other documents TBA

4 Work Breakdown Structure

4.1 Work Breakdown Structure

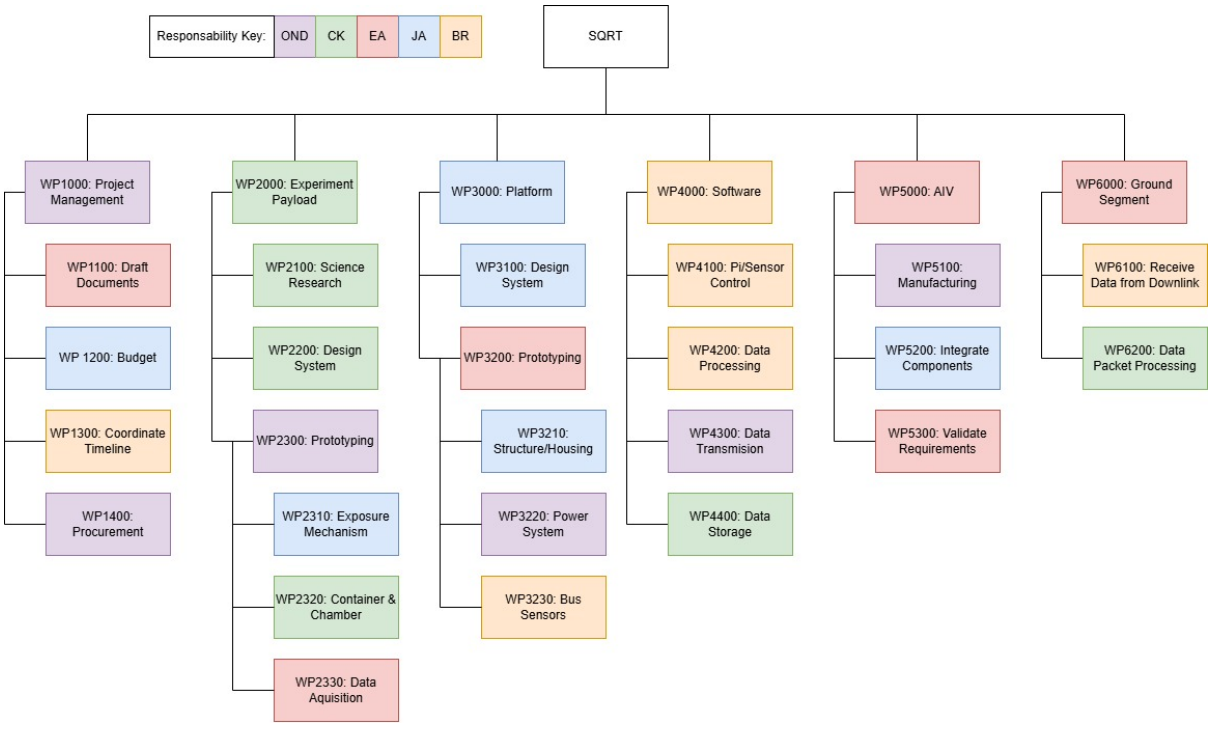


Figure 2: SQRT Work Breakdown Structure

The SQRT mission will follow the work breakdown structure shown in Figure 2 to ensure that all necessary work for the SQRT mission is completed within the schedule and all requirements detailed in documents SQT-PDR-MDD-0 and SQT-PDR-SEP-0 are met. While work on SQRT systems in collaborative, individual team members are assigned the final responsibility of ensuring completion of

individual work packages. Responsibility for signing off on individual work packages is assigned by color using the key in the upper left-hand corner of Figure 2. Leadership roles of subsections of the SQRT mission are assigned in Sec. 3: Organisational Structure following the same color coding. These subsections align with work packages WP1000, WP2000, WP3000, WP4000, and WP5000.

4.2 Work Packages

The following work packages found in Table 2 define the work that will be done to complete the SQRT mission requirements and achieve mission goals. Work packages are also represented on a Gantt chart timeline in Sec. 5.

Table 2: SQRT Work Packages

WP Lvl	WP Title	Task	Prerequisites & Dependencies	Deliverables	Start	End	Owner
1	WP1000	Complete the SQRT Project Management	WP1100, WP1200, WP1300, WP1400	PDR, CDR, FAR management deliverables	20/1	28/4	OND
2	WP1100	Draft Mission Documents	Mission concept, requirements	MDD, SEP, DJF, PMP Documents	20/1	12/4	EA
2	WP1200	Define and Track Project Budget	BOM, Procurement Plan	Approved and tracked budget	20/1	21/2	JA
2	WP1300	Coordinate Project Timeline	WBS, review milestones	Integrated project schedule	20/1	28/4	BR
2	WP1400	Procure Project Components	BOM, Budget approval	Procured hardware and materials	25/2	28/2	OND
1	WP2000	Complete Experiment payload	WP2100, WP2200, WP2300, WP5000	Functional experiment payload	19/1	3/4	CK
2	WP2100	Perform Science Research	Literature, mission goals	Scientific justification and models	19/1	27/2	CK
2	WP2200	Design Experiment Payload	Requirements, science results	Payload design ideas	26/1	27/2	CK
2	WP2300	Prototype Experiment Payload	WP2200	Payload prototype, completed WP2310, WP2320, WP2330.	24/2	6/3	OND
3	WP2310	Develop Exposure Mechanism	Payload design	Functional exposure mechanism	24/2	6/3	JA
3	WP2320	Develop Container and Chamber	Structural and thermal requirements	Container and chamber	24/2	6/3	CK
3	WP2330	Prototype Payload Data Acquisition System	Sensor requirements	Experiment data acquisition system	24/2	6/3	EA
1	WP3000	Develop Platform Subsystem	WP3100, WP3200, WP5000	Functional platform subsystem	26/1	3/4	JA

2	WP3100	Design Platform Architecture	System requirements	Platform design documentation, initial payload design ideas	2/2	27/2	JA
2	WP3200	Prototype platform	WP3100	Platform prototype, completed WP3310, WP3320, WP3330.	24/2	6/3	EA
3	WP3210	Prototype Structure and Housing	Structural design	Platform structure and housing	24/2	6/3	JA
3	WP3220	Create Power System	Power requirements	Operational power subsystem	24/2	19/3	OND
3	WP3230	Prototype Platform Sensors	Sensor specifications	Integrated platform sensors	9/3	19/3	BR
1	WP4000	Software	System requirements	Flight software suite	16/2	3/4	BR
2	WP4100	Develop Onboard Control Software	Pi/Sensor interfaces	Pi/Sensor control software	16/2	3/4	BR
2	WP4200	Develop Data Processing Software	Raw experiment data	Processed data products	9/3	3/4	BR
2	WP4300	Develop Data transmission software	Communications requirements	Data transmission software	9/3	3/4	OND
2	WP4400	Develop Data Storage Software	Storage Requirements	Onboard data storage system	16/2	9/3	CK
1	WP5000	Perform Assembly, Integration and Verification	WP5100, WP5200, WP5300	Integrated and verified system ready for FAR	9/3	3/4	EA
2	WP5100	Manufacture System Components	Final designs, WP2200, WP3200.	Manufactured Components	9/3	19/3	OND
2	WP5200	Integrate Subsystems	Completed subsystems, WP5100.	Integrated TupperSat	16/3	3/4	JA
2	WP5300	Validate Requirements via Testing	Integrated system, WP5200, WP4000.	Verified requirements	16/3	3/4	EA
1	WP6000	Operate Ground Segment	FAR complete, payload launch.	Ground segment operations WP6100, WP6200, WP6300.	14/4	25/4	EA
2	WP6100	Receive Data from Downlink	Receiver	Received flight data	14/4	25/4	BR
2	WP6200	Process Downlinked Data	Raw telemetry	Processed flight data	14/4	25/4	CK

5 Project Timeline

5.1 Gantt Chart

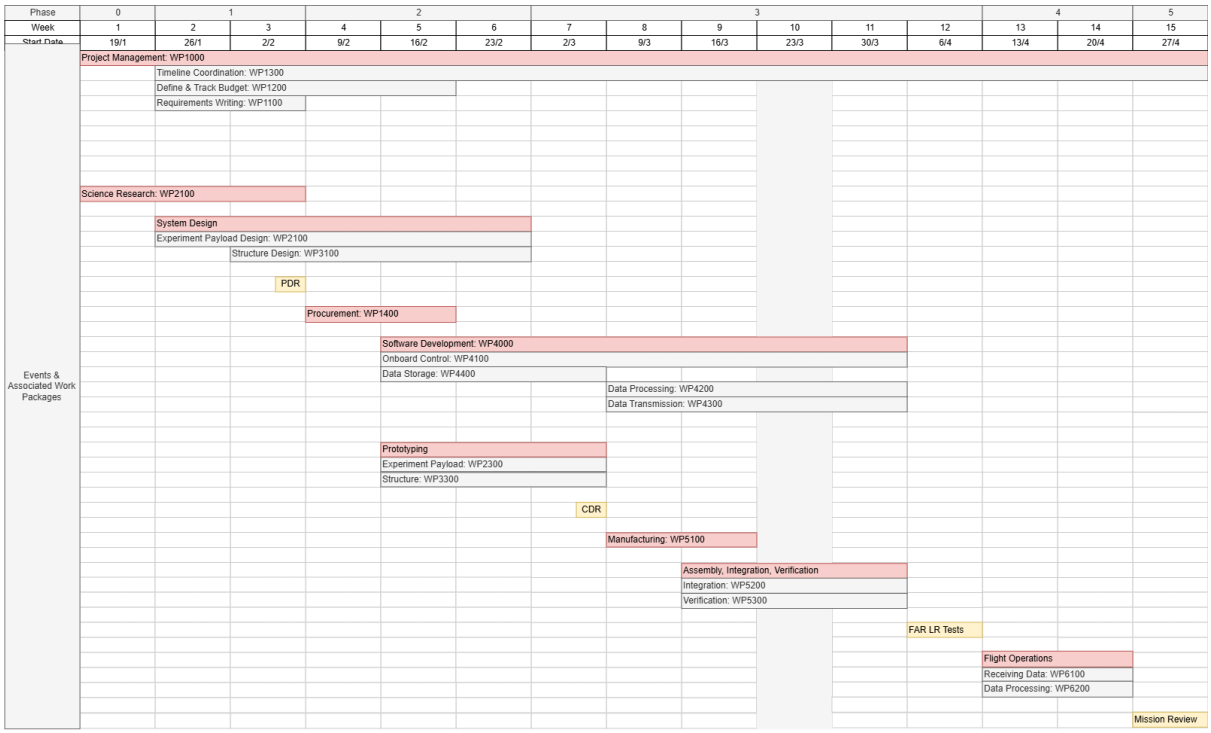


Figure 3: SQRT Mission Gantt Chart

The schedule our team will follow to develop, manufacture, test, and launch the SQRT mission is shown above. The grey column in week 10 indicates a week where the team will be away from the lab due to a program trip. The PDR, CDR, and FAR are highlighted in yellow. These milestones and other key dates are pulled from UCD-SCD-2026-0, and are described in Table 3. Related work packages are listed with tasks, and dependent work packages are shown in a light grey below parent tasks in red. Certain tasks can be started before earlier tasks are finished, such as prototyping which will start with in-house components before procurement is done but cannot be fully completed until after all the necessary products are acquired. The total duration of this mission will be 99 days from the kick-off meeting on 19 January 2026 to the feedback session with UCD on 28 April 2026. Mission close-out review reports (MCR) will be submitted on an individual basis on 19 May 2026.

Table 3: Key Mission Dates

Date	Event	Deliverables	Description
19 Jan. 2026	KOM	None	Project Kick Off
06 Feb. 2026	PDR	PDR Data Pack	Interview with UCD
27 Feb. 2026	CDR	CDR Data Pack	Interview with UCD
10 Apr. 2026	FAR	TupperSat	Presentation to UCD
14 Apr. 2026	Launch Window Opens	TupperSat	11 Day Launch Window
28 Apr. 2026	Mission Review & Feedback Session	Mission Analysis	Meeting with UCD to review objective completion
19 May 2026	MCR Submission	MCR	Individual Analysis

5.2 Mission Phases

Phase 0: The mission concept and goals are developed following the kick-off meeting.

Phase 1: An initial system design and project requirements are drafted, and the PDR documents are written. This phase ends with the Preliminary Design Review (PDR).

Phase 2: Following acceptance of the PDR, the design is matured and prototyping and software development work begins. Materials are procured and CDR documents are written. This phase ends with the Critical Design Review (CDR).

Phase 3: Following acceptance of the CDR, procurement, prototyping, and software development steps are completed. The final TupperSat design is manufactured and tested, including long range tests. This phase ends with the Flight Acceptance Review (FAR).

Phase 4: The SQRT TupperSat is launched and operates.

Phase 5: Data from the mission operation is analysed and the mission is reviewed.

6 Budget & Procurement

6.1 Initial Budget

Funding from University College Dublin for this mission is set to €100.00 with a contingency of €20.00 (total capped amount: €120). All necessary materials not provided by UCD will be procured using this funding. A list of the materials to be procured is shown in the bill of materials in Table 4. The budget for this mission is calculated based on the expected costs of items (and shipping where necessary). This list of materials is derived from the design overview in SQT-PDR-DJF-0. Items ordered from UCD suppliers Mouser and Farnell will not incur shipping costs for this mission. There are no labor costs associated with this project as all team members are unpaid students. The current budgeted expenses total €82.71, leaving a 17% margin (excluding the additional €20 contingency).

6.2 Bill of Materials

Table 4: Bill of Materials

Item	Descriptor	Amount	Source	Cost(in €)
RPI NoIR Camera Board	Camera to observe water flash during experiment.	2	Farnell Order Code: 3677846	20.64
Distilled Water	Distilled water will be used to reduce uncontrolled variables in the experiment.	1 L	Lidl	2.59
Tupperware Container	A Tupperware contained maximum 2L in volume will be used to encapsulate the SQRT payload. It will not be structural.	1	Lidl	7.00
Servo Motor	Used in valve mechanism for experiment payload.	1	Farnell Part No. SER0043	13.10
15 mL Syringe	A syringe will be modified and used as the water container.	5	firstaidshop.ie SKU: 15MLSINGLES	11.23
MSB5611-01BA03	Second pressure sensor for use inside the Low Pressure Chamber.	1	Mouser	8.00

DS18B20 digital thermometer	A second thermometer to be used inside the Low Pressure Chamber to verify the experiment mechanism functions.	1	Mouser	8.60
Electric Heating Pad	A Heating pad will be used to manage water container temperature.	1	The Pi Hut SKU: ADA1481	11.55
			Total	**82.71**

6.3 Procurement Plan

Procurement will begin in Week 4 and end in Week 5 (During Phase 2). The primary suppliers for electric components are Mouser and Farnell. The Mouser shipping time is quoted per their website is 1-3 business days and the Farnell shipping time is 6 days. The SQRT timeline has 2 weeks set for procurement to accommodate the order process and these shipping times.

Possible Cost-Saving Measures: It may be possible to source the distilled water and syringes from UCD labs, and the pressure and thermometer sensors from UCD Physics, for a lower cost.